

Project Initialization and Planning Phase

Date	29/09/2026
Team ID	SWTID-2026-1297
Project Title	AI FAQ Assistant API
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

To develop a centralized AI-powered FAQ and customer-support backend that allows users to register, authenticate, search FAQs, manage FAQ content, and generate FAQ content using Google Gemini AI.

Project Overview	
Objective	To develop a centralized AI-powered FAQ and customer-support backend that allows users to register, authenticate, search FAQs, manage FAQ content, and generate FAQ content using Google Gemini AI.
Scope	The project focuses on developing a REST API backend for FAQ and customer-support management, including user authentication, FAQ creation and management, FAQ search, validation, error handling, and AI-assisted FAQ generation.
Problem Statement	
Description: Users may face difficulty in finding relevant answers to frequently asked questions and obtaining quick customer-support information. A centralized API is required to manage FAQ content, provide search functionality, and support AI-assisted FAQ generation. Impact: The proposed system provides centralized FAQ management, easier FAQ retrieval, search functionality, and AI-assisted FAQ generation, helping improve the availability and accessibility of customer-support information.	
Proposed Solution	
Develop a modular REST API using Node.js and Express.js, with MongoDB for storing user and FAQ data, JWT for authentication, and Google Gemini AI for AI-powered	

FAQ generation.	
Key Features	• User registration and login • JWT authentication and protected routes • FAQ create, read, update and delete operations



	• .FAQ search and categorization • AI-powered FAQ generation using Gemini • Validation and centralized error handling
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Computer/Laptop for backend development and API testing	Not specifically specified
Memory	RAM required for development environment	Not specifically specified
Storage	Storage for project files, dependencies and database	Not specifically specified
Software		
Frameworks • Resource Type: Frameworks • Description: Backend/API framework • Specification/Allocation: Node.js, Express.js		
Libraries		

- Resource Type: Libraries
- Description: Database, authentication and AI libraries
- Specification/Allocation: Mongoose, bcrypt, JWT, Google Gemini AI

Development Environment

- Resource Type: Development Environment
- Description: IDE and API testing tools
- Specification/Allocation: Visual Studio Code, Postman / Thunder Client

Data	User and FAQ information	MongoDB database
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